

Operating Systems (OS)

Comprehensive Theory, System Concepts, Real-World Examples & 25 Practice Questions

1. Introduction to Operating Systems

An **Operating System (OS)** is system software that acts as an intermediary between the computer hardware and the user. Its primary goal is to execute user programs, make the computer system convenient to use, and utilize computer hardware in an efficient manner.

Core Functions of an Operating System

- **Process Management:** Allocates CPU time to active execution units (threads/processes).
- **Memory Management:** Tracks primary memory allocation and manages paging/virtual memory.
- **File System Management:** Manages directories, files, and disk access rights.
- **Device Management:** Communicates with hardware components using Device Drivers.
- **Security & Protection:** Guards resources against unauthorized software or user access.

2. Core Operating System Concepts & Examples

1. Types of Operating Systems

Operating systems have evolved into various types based on architecture and usage:

- **Batch OS:** Jobs with similar needs are batched together and executed sequentially without user interaction.
- **Time-Sharing / Multitasking OS:** CPU time is divided into small time slices (quantums) so multiple users/tasks run concurrently.
- **Real-Time OS (RTOS):** Guarantees strict timing constraints for task execution (Hard RTOS vs Soft RTOS).
- **Distributed OS:** Manages a collection of independent computers and presents them as a single system.

Real-World Example: An Air Traffic Control system uses a **Hard RTOS** where a delay in processing data can lead to catastrophic accidents, whereas a laptop running **Windows/macOS** uses a **Time-Sharing OS** to seamlessly run Spotify, Web Browsers, and Word Processors at once.

2. Process Management & CPU Scheduling

A process is a program in execution. The OS uses CPU Scheduling algorithms to decide which process gets CPU access:

- **First-Come, First-Served (FCFS):** Non-preemptive, executes processes in order of arrival.
- **Shortest Job First (SJF):** Schedules the process with the shortest execution time next.
- **Round Robin (RR):** Preemptive algorithm where each process receives a fixed time slice (quantum).
- **Priority Scheduling:** Assigns priorities to processes; highest priority executes first.

Real-World Example: A print queue uses **FCFS** (first document sent gets printed first), while your smartphone OS uses **Priority Scheduling** (an incoming phone call interrupts a running background game).

3. Process Synchronization & Deadlocks

When multiple processes share resources, synchronization mechanisms like **Semaphores** and **Mutexes** prevent race conditions.

A **Deadlock** occurs when a set of processes are blocked because each holds a resource and waits for another resource held by another process. The 4 necessary conditions for Deadlock are: *Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait*.

Real-World Example: Imagine a narrow one-lane bridge where two cars meet from opposite directions. Neither can move forward without the other backing up — this represents a **Deadlock** state.

4. Memory Management & Virtual Memory

Techniques used by OS to manage main memory (RAM):

- **Paging:** Memory management scheme that stores/retrieves data from secondary storage in fixed-size blocks called *pages* (main memory is divided into *frames*).
- **Segmentation:** Memory scheme that divides memory into logical variable-length segments (code, data, stack).
- **Virtual Memory:** Allows execution of processes that are larger than physical RAM by swapping pages to secondary storage (Page Fault handling & Demand Paging).

Real-World Example: Launching a 16GB video game on a computer with only 8GB RAM works because the OS uses **Virtual Memory**, keeping only active game sections in RAM while keeping the rest on the SSD page file.

5. Disk Management & File Systems

The OS manages storage structures using disk scheduling algorithms (SSTF, SCAN, C-SCAN) and maintains file allocation tables (FAT32, NTFS, ext4) to organize files into hierarchical directories.

Real-World Example: When saving a document, the **File System** translates the file path `C:\Documents\report.pdf` into exact physical sector coordinates on your disk drive.

3. Additional Important Concepts

- **System Calls:** Programmatic requests made by a process to request a service from the OS kernel (e.g., `fork()`, `exec()`, `open()`, `read()`).
- **Kernel Modes:** Execution modes providing security boundary — *User Mode* (restricted) vs *Kernel/Supervisor Mode* (unrestricted hardware access).
- **Thrashing:** A severe condition where the OS spends more time swapping pages in and out of memory than executing actual process instructions.

4. Operating Systems Practice Questions (25 MCQs)

1. What is the main core module of an operating system called?

- | | |
|-------------------|-------------|
| A) Shell | B) Kernel |
| C) Command Prompt | D) Compiler |

2. Which CPU scheduling algorithm allocates a fixed time slice (quantum) to each process?

- A) First-Come First-Served
- B) Round Robin
- C) Shortest Job First
- D) Priority Scheduling

3. Which of the following is NOT one of the four necessary conditions for a deadlock?

- A) Mutual Exclusion
- B) Circular Wait
- C) Preemption allowed
- D) Hold and Wait

4. A program in execution is known as a:

- A) Instruction
- B) Process
- C) Task Manager
- D) System Call

5. In memory management, fixed-size blocks of main memory are called:

- A) Pages
- B) Frames
- C) Segments
- D) Sectors

6. Which system call is used in Unix-like systems to create a new child process?

- A) create()
- B) fork()
- C) new()
- D) spawn()

7. What condition occurs when the OS spends more time paging than executing instructions?

- A) Starvation
- B) Thrashing
- C) Deadlock
- D) Aging

8. Which Operating System type guarantees execution within strict time limits?

- A) Batch OS
- B) Real-Time OS (RTOS)
- C) Time-Sharing OS
- D) Network OS

9. Banker's Algorithm is used in Operating Systems for:

- A) Deadlock Recovery
- B) Deadlock Avoidance
- C) CPU Scheduling
- D) Page Replacement

10. What is Belady's Anomaly?

- A) Increasing page frames causes more page faults in FIFO
- B) CPU utilization reaches 100%
- C) Process priority is automatically lowered
- D) Disk speed reduces over time

11. What provides an interface between a running program and the OS kernel?

- A) System Calls
- B) Interrupt Vector
- C) GUI Shell
- D) Library Function

12. Which page replacement algorithm replaces the page that will not be used for the longest period of time?

- A) FIFO
- B) Optimal Page Replacement
- C) LRU
- D) LFU

13. What is a Semaphore in process synchronization?

- A) A hardware register
- B) An integer variable used for signaling/synchronization
- C) A type of disk sector
- D) A network protocol

14. Which memory allocation technique splits physical memory into variable-sized logical units?

- A) Paging
- B) Segmentation
- C) Formatting
- D) Spooling

15. What is the process state when a process is waiting to be assigned to a processor?

- A) Running
- B) Ready
- C) Blocked
- D) Terminated

16. What is a lightweight process that shares code, data, and resources with other units of the same process?

- A) Kernel
- B) Thread
- C) Semaphore
- D) Driver

17. SPOOLING stands for:

- A) Simultaneous Peripheral Operations On-Line
- B) Sequential Program Output Online
- C) System Performance Optimization Online
- D) Shared Process Execution On-Line

18. The technique of gradually increasing the priority of processes that wait for a long time is called:

- A) Paging
- B) Aging
- C) Swapping
- D) Dispatching

19. Which disk scheduling algorithm services requests closest to the current head position?

- A) FCFS
- B) SSTF (Shortest Seek Time First)
- C) SCAN
- D) LOOK

20. Virtual memory is usually implemented by which mechanism?

- A) Compaction
- B) Demand Paging
- C) Static Relocation
- D) Partitioning

21. What happens when a hardware device needs attention from the CPU?

- A) System Call
- B) Interrupt
- C) Page Fault
- D) Polling

22. What data structure stores complete information about a specific process (PID, state, registers)?

- A) Inode
- B) PCB (Process Control Block)
- C) Segment Table
- D) Page Directory

23. Translation Lookaside Buffer (TLB) is used as a fast cache for:

- A) Disk Sectors
- B) Page Table Entries
- C) Process IDs
- D) I/O Commands

24. What is the main cause of External Fragmentation?

- A) Fixed size pages
- B) Variable size free memory blocks separated by allocated blocks
- C) Internal stack overflow
- D) Slow hard drive access

25. Which component of the OS selects a process from the ready queue to execute on the CPU?

- A) Long-term Scheduler
- B) Short-term Scheduler (CPU Scheduler)
- C) Medium-term Scheduler
- D) Dispatcher

Answer Key

- | | | | | |
|------|-------|-------|-------|-------|
| 1. B | 6. B | 11. A | 16. B | 21. B |
| 2. B | 7. B | 12. B | 17. A | 22. B |
| 3. C | 8. B | 13. B | 18. B | 23. B |
| 4. B | 9. B | 14. B | 19. B | 24. B |
| 5. B | 10. A | 15. B | 20. B | 25. B |